Oncor Skyview 345/138-kV Switch Project – ERCOT Independent Review (EIR) Status Update

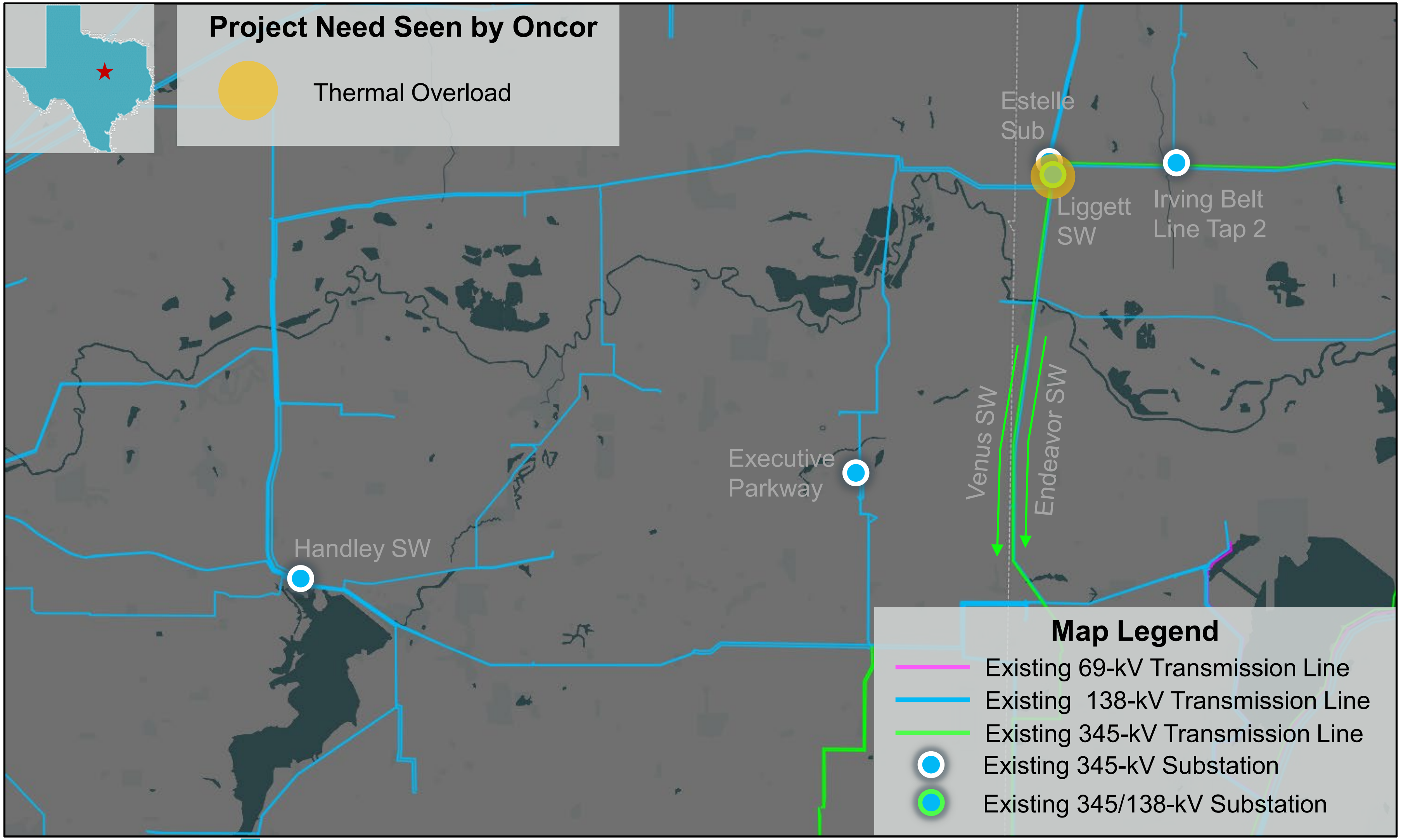
Tanzila Ahmed

RPG Meeting
March 17, 2026

Introduction

- Oncor submitted the Skyview 345/138-kV Switch Project for Regional Planning Group (RPG) review in September 2025
  - This Tier 2 project is estimated to cost approximately $56.82 million and will require a Certificate of Convenience and Necessity (CCN) filing
  - Estimated in-service date (ISD) is May 2027
  - Addresses the thermal overloads and voltage violations due to proposed load additions in the Dallas and Tarrant Counties
- Oncor provided an overview presentation and ERCOT provided the study scope at the October RPG meeting
  - https://www.ercot.com/calendar/10282025-RPG-Meeting
- ERCOT provided status update at the December 2025 and January 2026 RPG meetings
  - https://www.ercot.com/calendar/12162025-RPG-Meeting
  - https://www.ercot.com/calendar/01162026-RPG-Meeting
- This project is currently under ERCOT Independent Review (EIR)

Map with Project Need Seen by Oncor



Study Assumptions

- Study Region
  - North Central Weather Zone, focusing on the transmission elements in the Dallas and Tarrant counties
- Steady-State Base Case
  - Summer Peak: Final 2024RTP_2028_SUM_12202024
- Transmission (based on June 2025 TPIT report)
  - See Appendix A1 for the list of transmission projects that were added
  - See Appendix A2 for the list of 2024 RTP placeholder projects that was removed
- Generation updates (based on August 2025 GIS report)
  - See Appendix B for the list of generation projects that were added
- Load level in the study area was maintained consistent with the Final RTP case
- The reserve was maintained consistent with the 2024 RTP

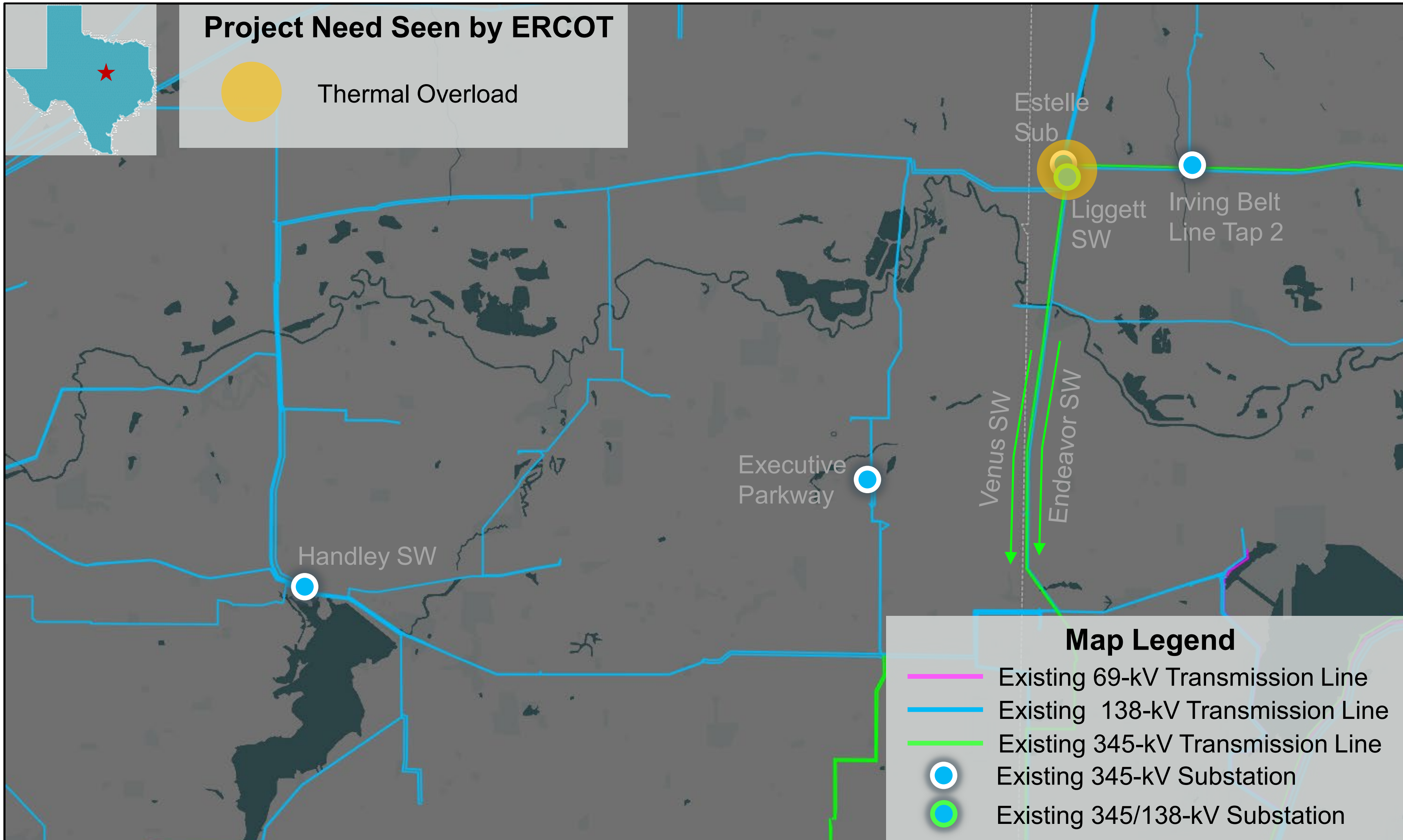
Results of Reliability Assessments

- ERCOT conducted steady-state load flow analysis for the study base case according to the NERC TPL-001-5.1 and ERCOT Planning Criteria to identify project need

Contingency Category	Voltage Violations	Thermal Violations	Unsolved Power Flow
N-0 (P0)	None	None	None
N-1 (P1, P2-1, P7)	None	None	None
G-1+N-1 (P3)*	None	None	None
X-1+N-1 (P6-2)*	None	2	None

* See Appendix C for list of G-1 generators and X-1 transformers tested

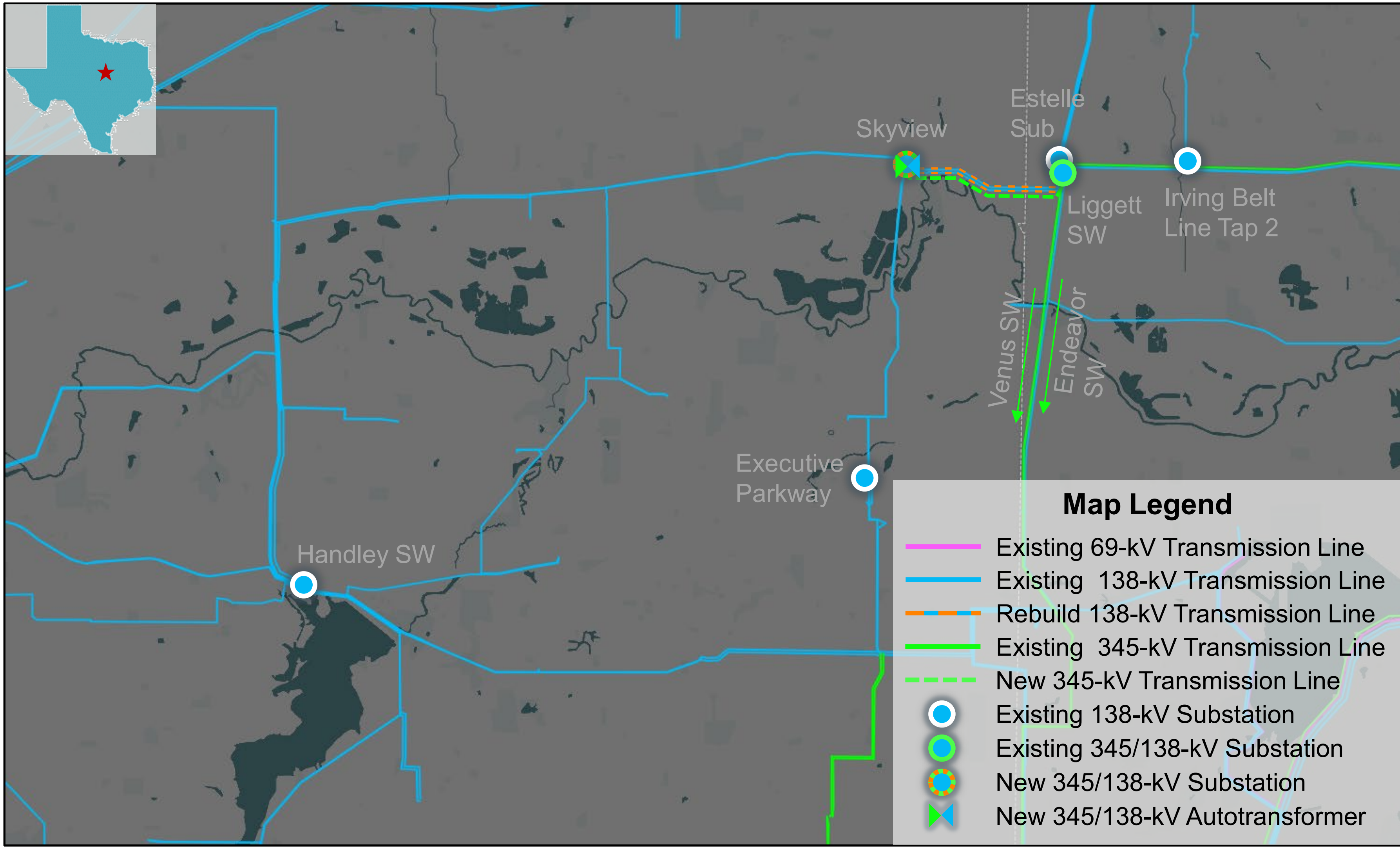
Study Area Map with Violations Seen by ERCOT



Option 1 – Oncor Proposed Option

- Construct a new Skyview 345/138-kV Switch (SW) by installing one 345-kV, 3,200 A breaker and seven 138-kV, 3,200 A breakers in a breaker-and-a-half bus arrangement
- Install one 600 MVA 345/138-kV Autotransformer at Skyview Switch with a normal and emergency ratings of 700 MVA and 750 MVA, respectively
- Rebuild portion of the existing Liggett Switch to Handley Switch / Executive Parkway 138-kV double-circuit transmission line on triple-circuit capable structures with both circuit in place, with normal and emergency ratings of at least 614 MVA, for approximately 2.6 miles/circuit
- Construct a new Liggett Switch to Skyview Switch 345-kV transmission line on the newly rebuild triple-circuit capable structures on the upgraded 2.6-mile portion of the exiting Liggett Switch to Handley Switch / Executive Parkway 138-kV double-circuit transmission line, with normal and emergency ratings of at least 1,912 MVA, for approximately 2.6 miles/circuit
- Install two 345-kV, 3,200 A circuit brakers at Liggett 345-kV Switch
- Ensure the 345-kV and 138-kV terminal equipment meet or exceed a rating of 3,200 A and 3,000 A, respectively

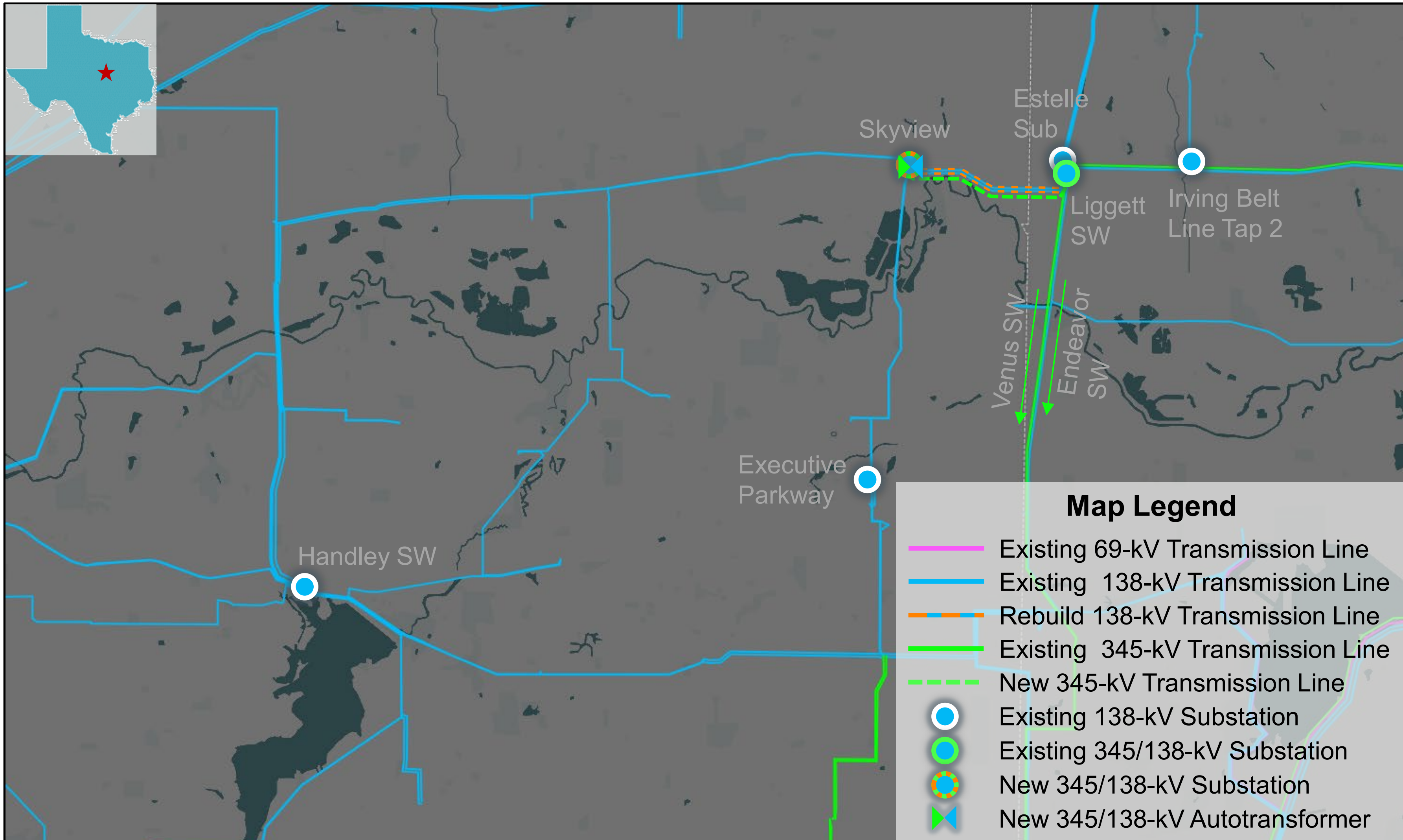
Option 1 – Oncor Proposed Option



Option 2 – Modified Oncor Proposed Option

- Construct a new Skyview 345/138-kV Switch (SW) by installing one 345-kV, 3,200 A breaker and seven 138-kV, 3,200 A breakers in a breaker-and-a-half bus arrangement
- Install one 600 MVA 345/138-kV Autotransformer at Skyview Switch with a normal and emergency ratings of 700 MVA and 750 MVA, respectively
- Rebuild portion of the existing Liggett Switch to Handley Switch / Executive Parkway 138-kV double-circuit transmission line on triple-circuit capable structures with both circuit in place, with normal and emergency ratings of at least 614 MVA, for approximately 2.6 miles/circuit
- Construct a new Liggett Switch to Skyview Switch 345-kV transmission line on the newly rebuild triple-circuit capable structures on the upgraded 2.6-mile portion of the exiting Liggett Switch to Handley Switch / Executive Parkway 138-kV double-circuit transmission line, with normal and emergency ratings of at least 1,912 MVA, for approximately 2.6 miles/circuit
- Install two 345-kV, 3,200 A circuit brakers at Liggett 345-kV Switch
- Ensure the 345-kV and 138-kV terminal equipment meet or exceed a rating of 3,200 A and 3,000 A, respectively
- Rebuild the existing Liggett Switch to Estelle 138-kV transmission line on double-circuit capable with only one circuit in place, with normal and emergency ratings of at least 495 MVA, for approximately 0.6 miles/circuit
- Rebuild the existing Liggett Switch to IRVVALVW 138-kV transmission line on double-circuit capable with only one circuit in place, with normal and emergency ratings of at least 495 MVA, for approximately 1.5 miles/circuit

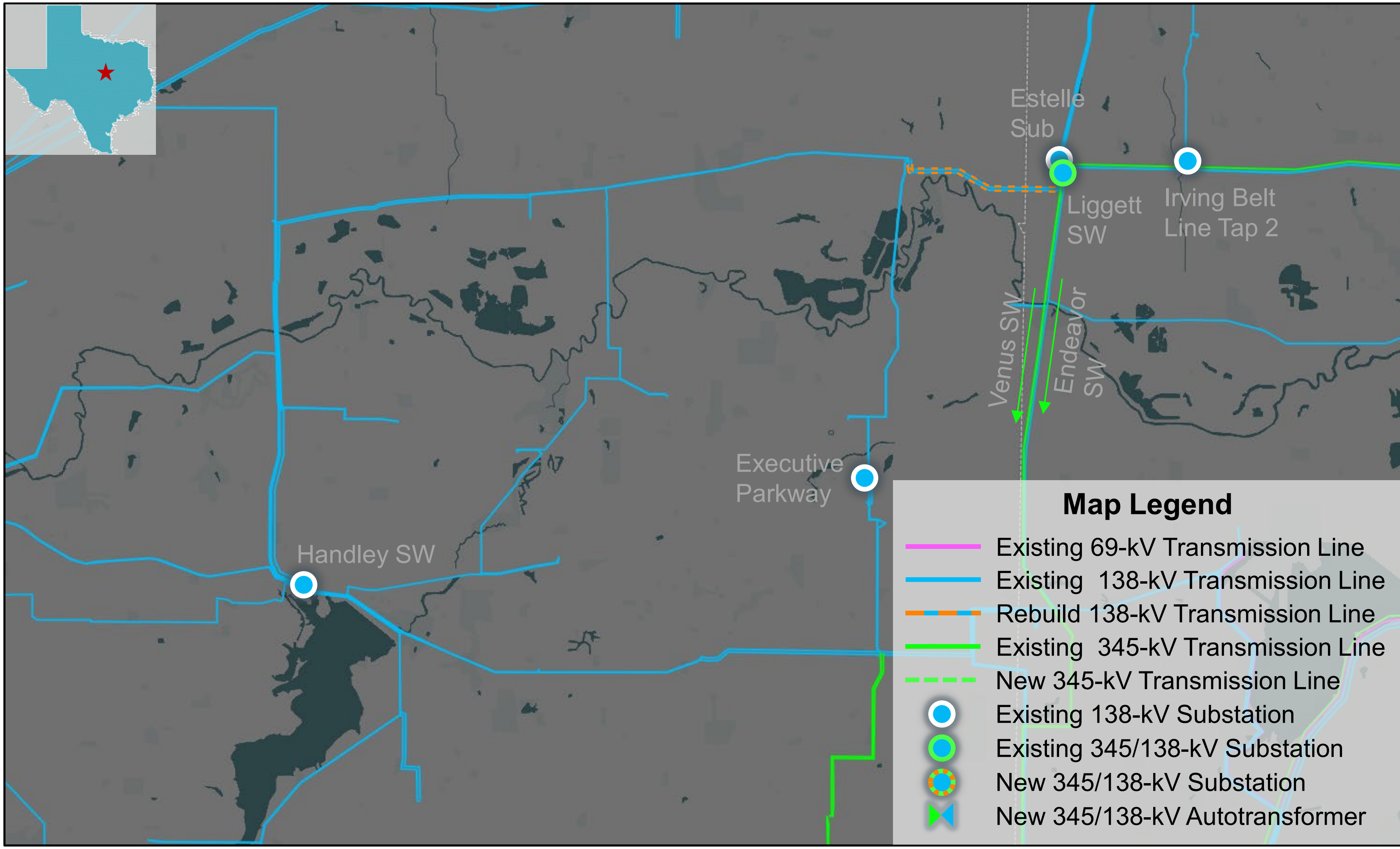
Option 2 – Modified Oncor Proposed Option



Option 3 – ERCOT Option

- Install a third new 600 MVA 345/138-kV Autotransformer at Liggett Switch with a normal and emergency ratings of 700 MVA and 750 MVA, respectively
- Rebuild portion of the existing Liggett Switch to Handley Switch / Executive Parkway 138-kV double-circuit transmission line on double-circuit capable structures with both circuit in place, with normal and emergency ratings of at least 614 MVA, for approximately 2.6 miles/circuit
- Rebuild the existing Liggett Switch to Estelle 138-kV transmission line on double-circuit capable with only one circuit in place, with normal and emergency ratings of at least 495 MVA, for approximately 0.6 miles/circuit
- Rebuild the existing Liggett Switch to IRVVALVW 138-kV transmission line on double-circuit capable with only one circuit in place, with normal and emergency ratings of at least 495 MVA, for approximately 1.5 miles/circuit

Option 3 – ERCOT Option



Results of Option Evaluation

- ERCOT conducted steady-state load flow analysis for the study base case according to the NERC TPL-001-5.1 and ERCOT Planning Criteria to evaluate options

Option	N-1		G-1*+N-1		X-1*+N-1		Unsolved Powerflow
	Thermal Violations	Voltage Violations	Thermal Violations	Voltage Violations	Thermal Violations	Voltage Violations	
1	None	None	None	None	None	None	None
2	None	None	None	None	None	None	None
3	None	None	None	None	2	None	None

* See Appendix C for list of G-1 generators and X-1 transformers tested

- Option 1 and Option 2 were short-listed for further evaluation

Results of Maintenance Outage Evaluation

- ERCOT conducted maintenance outage analysis on all short-listed options to compare relative performance of the options
  - Load levels the North Central Weather Zone were scaled down based on the historical non-summer peak data to 82%, in order to mimic the non-summer peak load condition
  - Based on the review of system topology of the area, ERCOT tested N-2 contingency combinations, and then tested all applicable contingency violations with system adjustments (N-1-1)
- No thermal or voltage constraints were observed in the N-1-1 analysis

Option	Thermal Violation	Voltage Violation
1	None	None
2	None	None

Long-Term Load-Serving Capability Assessment

- Assumptions
  - Adjusted load up in the study area, excluding Flexible Loads in the area
  - Adjusted conforming load down outside of the North Central Weather Zone to balance power
  - Based on N-1 contingency
- Preliminary Findings
  - Both of Option 1 and Option 2 were observed to have similar incremental load-serving capability

Option	Incremental Load-Serving Capability (~MW)
1	255
2	265

Cost Estimate and Feasibility Assessment

- Transmission Service Providers (TSPs) performed feasibility assessments and provided cost estimates for the two short-listed options

Option	Cost Estimates (~$M)*	CCN Required (~miles)	Feasibility	Expected ISD
1	56.8	Yes (2.6)	Yes	May 2027
2	57.4	Yes (2.6)	Yes	May 2027

* Cost estimate was provided by the TSPs and includes the estimated capital cost with energized construction work.

Comparison of Short-Listed Options

	Option 1	Option 2
Meets ERCOT and NERC Reliability Criteria	Yes	Yes
Improves Long-Term Load-Serving Capability (~MW)	255	265
Requires CCN (~miles)	Yes (2.6)	Yes (2.6)
Cost Estimate* (~$M)	56.8	57.4
Feasible	Yes	Yes
Expected ISD	May 2027	May 2027

* Cost estimate was provided by the TSP(s) and includes the estimated capital cost with energized construction work.

- Both options have similar long-term load-serving capability
- Option 1 is the least cost option

ERCOT Preferred Option

- Option 1 was selected as the preferred option because it
  - Addresses the project needs and meets both ERCOT and NERC reliability criteria
  - Is the least cost solution
  - Improves long-term load-serving capability for future load growth in the area

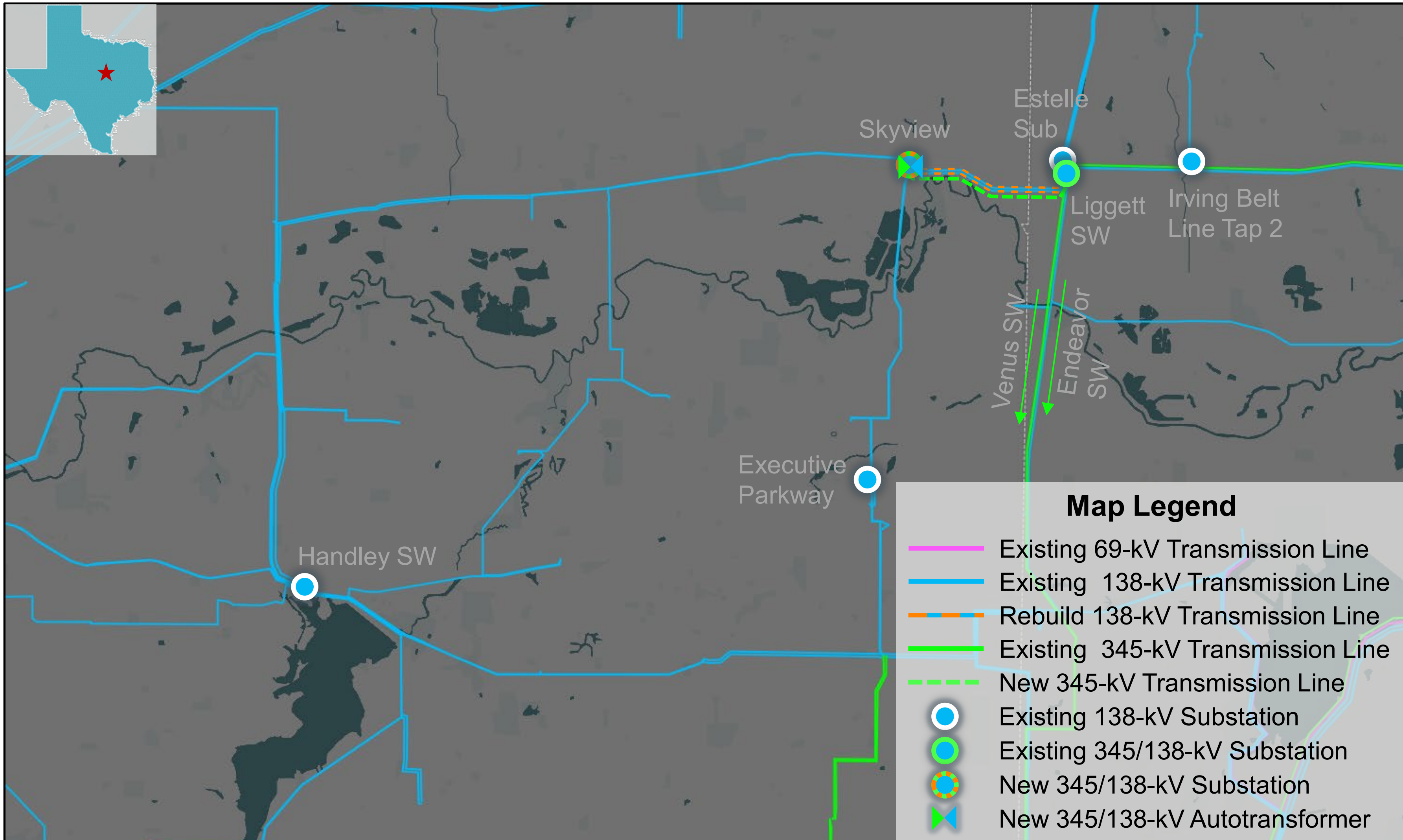
Additional Analysis

- Congestion analysis was performed for the preferred option using the 2024 RTP 2029 economic case
  - The preferred option did not result in significant new congestion within the study area

ERCOT Recommendation

- ERCOT recommends Option 1
  - Estimated Cost: approximately $56.82 million
  - Expected ISD: May 2027
  - CCN filling will be required to
    - Construct a new Liggett Switch to Skyview Switch 345-kV transmission line on the newly rebuilt triple-circuit capable structures on the upgraded 2.6 miles portion of the exiting Liggett Switch to Handley Switch / Executive Parkway 138-kV double-circuit transmission line

Map of ERCOT Recommended Option



ERCOT Recommended Option

- Construct a new Skyview 345/138-kV Switch (SW) by installing one 345-kV, 3,200 A breaker and seven 138-kV, 3,200 A breakers in a breaker-and-a-half bus arrangement
- Install one 600 MVA 345/138-kV Autotransformer at Skyview Switch with a normal and emergency ratings of 700 MVA and 750 MVA, respectively
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- Install two 345-kV, 3,200 A circuit brakers at Liggett 345-kV Switch
- Ensure the 345-kV and 138-kV terminal equipment meet or exceed a rating of 3,200 A and 3,000 A, respectively

Next Steps

- Tentative timeline
  - EIR report to be posted in the MIS in March 2026

Thank you!

Stakeholder comments also welcomed through:

Tanzila.Ahmed@ercot.com

Robert.Golen@ercot.com

Appendix A1 – Transmission Projects to Add

- List of transmission projects to be added to study base case

RPG/TPIT No	Project Name	Tier	Project ISD	From County
24RPG007	Canton Area Loop Project	Tier 2	10/1/2026	Van Zandt
24RPG016	Rand Area Loop Project	Tier 2	4/1/2027	Kaufman
24RPG017	Venus Switch to Sam Switch 345-kV Line Project	Tier 1	5/1/2026	Ellis, Hill
24RPG021	Forney 345/138-kV Switch Rebuild Project	Tier 1	12/1/2025	Kaufman
24RPG022	Wilmer 345/138-kV Switch Project	Tier 1	5/1/2026	Dallas

Appendix A2 – Transmission Project to be Removed

- List of placeholder transmission project that is directly related to the proposed project were removed from the study base case

RTP Project ID	Project Name	County(s)
2024-NC77	Skyview (2231) 345/138-kV Substation Addition and Area Upgrade	Dallas, Tarrant

Appendix B – New Generation Projects to Add

GINR	Project Name	County	Projected COD	Fuel	Max Capacity (~MW)
21INR0359	Hickerson Solar	Bosque	11/21/2025	SOL	311.1
22INR0437	TORMES SOLAR	Navarro	3/31/2027	SOL	356.6
23INR0266	Langer Storage	Bosque	3/1/2027	OTH	61.5
23INR0538	Roadrunner Crossing BESS SLF	Eastland	2/13/2026	OTH	182.6
24INR0136	Eagle Springs Storage	Delta	12/31/2026	OTH	33.1
24INR0137	Eagle Springs Solar	Delta	12/31/2026	SOL	77.2
24INR0181	Bynum Solar Project	Coryell	1/21/2026	SOL	56.0
24INR0188	Tehuacana Creek Solar	Navarro	3/10/2027	SOL	505.4
24INR0189	Tehuacana Creek BESS SLF	Navarro	3/10/2027	OTH	0.0
24INR0331	Chisme Storage	Brown	4/13/2027	OTH	146.0
24INR0333	Chisme Solar	Brown	4/13/2027	SOL	147.0
24INR0364	Pitts Dudik II	Hill	2/27/2026	SOL	30.2
24INR0631	Radian Storage SLF	Brown	11/7/2025	OTH	0.0
25INR0018	Yellow Cat Wind	Navarro	4/1/2027	WIN	262.0
25INR0101	Drake BESS	Collin	7/14/2026	OTH	257.3
25INR0166	Padrino Solar	Hill	7/30/2026	SOL	201.4
25INR0199	Bonham Solar SLF	Limestone	8/31/2026	SOL	138.4
25INR0229	OCI Cobb Creek Solar	Hill	12/1/2026	SOL	203.1
25INR0231	Apache Hill BESS	Hood	7/31/2026	OTH	201.2
25INR0233	OCI Cobb Creek ESS	Hill	12/1/2026	OTH	201.6
25INR0391	Purple Sage BESS 1	Collin	5/30/2027	OTH	156.0
25INR0392	Purple Sage BESS 2	Collin	5/30/2027	OTH	156.0

Appendix C – G-1 Generators and X-1 Transformers

G-1 Generators	X-1 Transformers
Forney Combine Cycle Train	Liggett Switch 345/138-kV Ckt 1
Handly SES	Liggett Switch 345/138-kV Ckt 1
Mt. Creek SES	Miller Road Switch 345/138-kV
	Norwood Switch 345/138-kV
	Venus Kemp Ranch Switch 345/138-kV